

SOLIDWORKS API (.NET VBA C#) — Reference

(306/337) ISwPEToken- ISymmetricMateFeatureData

SOLIDWORKS API Documentation (converted from CHM)

ISwPEToken (Type)

ISwPEToken Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISwPEToken Interface

For future use

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwPEToken
```

Visual Basic (Usage)

```
Dim instance As ISwPEToken
```

C#

```
public interface ISwPEToken
```

C++/CLI

```
public interface class ISwPEToken
```

Visual Basic for Applications (VBA) Syntax

See [SwPEToken](#).

Accessors

[ISwPEClassFactory::SetPartnerKey](#)

See Also

[ISwPEToken Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISwPEToken Interface Members

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISwPEToken Interface

The following tables list the members exposed by [ISwPEToken](#).

See Also

[ISwPEToken Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISWPropertySheet (Type)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISWPropertySheet Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISWPropertySheet Interface

Allows applications to add pages to certain property sheets that are exported by the SOLIDWORKS Design application.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISWPropertySheet
```

Visual Basic (Usage)



```
Dim instance As ISWPropertySheet
```

C#

```
public interface ISWPropertySheet
```

C++/CLI

```
public interface class ISWPropertySheet
```

Visual Basic for Applications (VBA) Syntax

See [SWPropertySheet](#).

Remarks

See [swPropSheetType_e](#) for a list of the property sheets that support this functionality.

An example of a property sheet is the dialog that is displayed when you click **Tools > Options**. This dialog consists of a base property sheet and property pages stacked on top of it. To display a specific property page, click its tab.

The MFC class that implements a property sheet is CPropertySheet, and the MFC class that implements a property page is CPropertyPage. ISWPropertySheet is a OLE wrapper class over the SOLIDWORKS CPropertySheet class. This means that SOLIDWORKS API programmers can add their own CPropertyPages to a SOLIDWORKS Design property sheet.

A SOLIDWORKS Design property sheet is different from an [IPropertyManager](#) page. A PropertyManager page is displayed when the PropertyManager tab is selected on the left-side panel. PropertyManager pages have replaced numerous dialogs.

Creating PropertyPages

When a property sheet is created that can have pages added to it, the notification swAppPropertySheetCreateNotify is sent via the [ISldWorks](#) interface. This notification is sent as swAppPropertySheetCreateNotify(LPDISPATCH sheet, long type) where sheet is the IDispatch interface of a ISWPropertySheet object and type is an enumerated constant of type swPropSheetType_e.

When the application receives this notification, it can check the type of the sheet from the type argument. If pages are to be added, then the sequence of operations is:

- Ensure that the application resources will be used to make the page (for example by calling AfxSetResourceHandle())
- Create the CPropertyPage-derived class
- For 32-bit applications, pass the CPropertyPage object to [ISWPropertySheet::AddPage](#), casting the CPropertyPage* to a long
- For 64-bit applications, pass the CPropertyPage object to [ISWPropertySheet::AddPagex64](#), casting the CPropertyPage* to a long long
- Reset the resources to their previous state

Destroying PropertyPages

The CPropertyPage PostNcDestroy function is called by the SOLIDWORKS Design application when the property sheet is being destroyed. At this point, the application should delete any objects (for example the CPropertyPage object.)

Events are implemented with delegates in the Microsoft .NET Framework. See the [Overview](#) topic for a list of delegates for this interface.

Access Diagram

See Also

[ISWPropertySheet Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISWPropertySheet Interface Members

[See Also](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISWPropertySheet Interface

The following tables list the members exposed by [ISWPropertySheet](#).

Public Methods

Name	Description
 AddActivePage	Adds a third-party CPropertyPage to ISWPropertySheet and adds an ActiveX control on top of the page.
 AddPage	Obsolete. Superseded by ISWPropertySheet::AddPage64. This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016. The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used. Adds a CPropertyPage-derived page to an ISWPropertySheet .
 AddPage64	Adds a CPropertyPage-derived page to an ISWPropertySheet in 64-bit applications.
 GetControl	Gets the ActiveX control on the property sheet.
 RemovePage	Removes a CPropertyPage derived page from an ISWPropertySheet .

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See Also

[ISWPropertySheet Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISWPropertySheet Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISWPropertySheet Interface

For a list of all members of this type, see [ISWPropertySheet members](#).

Public Methods

Name	Description
 AddActivePage	Adds a third-party CPropertyPage to ISWPropertySheet and adds an ActiveX control on top of the page.
 AddPage	Obsolete. Superseded by ISWPropertySheet::AddPage64. This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016. The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used. Adds a CPropertyPage-derived page to an ISWPropertySheet .
 AddPage64	Adds a CPropertyPage-derived page to an ISWPropertySheet in 64-bit applications.
 GetControl	Gets the ActiveX control on the property sheet.
 RemovePage	Removes a CPropertyPage derived page from an ISWPropertySheet .

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See Also

[ISWPropertySheet Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

AddActivePage Method (ISWPropertySheet)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISWPropertySheet Interface](#) : AddActivePage Method (ISWPropertySheet)

Adds a third-party CPropertyPage to [ISWPropertySheet](#) and adds an ActiveX control on top of the page.

.NET Syntax

Visual Basic (Declaration)

```
Function AddActivePage( _  
    ByVal Title As System.String, _  
    ByVal ProgId As System.String, _  
    ByVal LicenseKey As System.String _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISWPropertySheet  
Dim Title As System.String  
Dim ProgId As System.String  
Dim LicenseKey As System.String  
Dim value As System.Integer
```

```
value = instance.AddActivePage(Title, ProgId, LicenseKey)
```

C#

```
System.int AddActivePage(  
    System.string Title,  
    System.string ProgId,  
    System.string LicenseKey  
)
```

C++/CLI

```
System.int AddActivePage(  
& System.String^ Title,  
& System.String^ ProgId,  
& System.String^ LicenseKey  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Title

Title of CPropertyPage

ProgId

Name, ProgID, or CLSID of the ActiveX control (see **Remarks**)

LicenseKey

License key for the ActiveX control

Return Value

Index of CPropertyPage on ISWPropertySheet

Visual Basic for Applications (VBA) Syntax

See [SWPropertySheet::AddActivePage](#).

Example

'VBA Main module

```
Dim swApp As SldWorks.SldWorks  
Option Explicit  
Sub main()  
    UserForm1.Show (swModeLess)  
End Sub
```

```
'Insert UserForm module to VBA application
```

```
Option Explicit
```

```
Public WithEvents swAppEvents As SldWorks.SldWorks
```

```
Dim swApp As SldWorks.SldWorks
```

```
Private Function swAppEvents_PropertySheetCreateNotify(ByVal Sheet As Object, ByVal sheetType As Long) As Long
```

```
    Dim msrcSWPropertySheet As SWPropertySheet
```

```
    Set msrcSWPropertySheet = Sheet
```

```
    Call msrcSWPropertySheet.AddActivePage("PropertySheetExample2", "", "")
```

```
End Function
```

```
Private Sub UserForm_Initialize()
```

```
    Set swApp = Application.SldWorks
```

```
    Set swAppEvents = swApp
```

```
End Sub
```

Remarks

Typically, this method is called from the ISldWorks [PropertySheetCreateNotify](#) event handler. See the Example section.

The ProgID argument accepts any of these values:

- OLE short name or ProgID for the class; for example, "MSCAL.Calendar". The name must match the name registered by the control.
- String form of a CLSID, contained within braces, for example, "{9DBAFCCF-592F-101B-85CE-00608CEC297B}".

See also [Keystrokes and Accelerator Keys in ActiveX Modeless Dialogs and PropertyManager Pages](#).

See Also

[ISWPropertySheet Interface](#)

[ISWPropertySheet Members](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddPage Method (ISWPropertySheet)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISWPropertySheet Interface](#) : AddPage Method (ISWPropertySheet)

Obsolete. Superseded by [ISWPropertySheet::AddPage64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Adds a CPropertyPage-derived page to an [ISWPropertySheet](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddPage( _  
    ByVal Page As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISWPropertySheet  
Dim Page As System.Integer  
Dim value As System.Integer
```

```
value = instance.AddPage(Page)
```

C#



```
System.int AddPage(  
    System.int Page  
)
```

C++/CLI

```
System.int AddPage(  
& System.int Page  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Page

Pointer to the CPropertyPage to add, cast to a long

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SWPropertySheet::AddPage](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

See Also

[ISWPropertySheet Interface](#)
[ISWPropertySheet Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddPagex64 Method (ISWPropertySheet)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISWPropertySheet Interface](#) : AddPagex64 Method (ISWPropertySheet)

Adds a CPropertyPage-derived page to an [ISWPropertySheet](#) in 64-bit applications.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPagex64( _  
    ByVal Page As System.Long _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISWPropertySheet  
Dim Page As System.Long  
Dim value As System.Integer  
  
value = instance.AddPagex64(Page)
```

C#

```
System.int AddPagex64(  
    System.long Page  
)
```

C++/CLI

```
System.int AddPagex64(  
& System.int64 Page  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Page
Pointer to the CPropertyPage to add, cast to a long long

Return Value

True if successful, false if not

Remarks

This method is only available through early binding and with 64-bit versions of the SOLIDWORKS Design software.

See Also

[ISWPropertySheet Interface](#)
[ISWPropertySheet Members](#)
[ISWPropertySheet::AddPage Method](#)

Availability

SOLIDWORKS 2010 SP01, Revision Number 18.1

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetControl Method (ISWPropertySheet)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISWPropertySheet Interface](#) : GetControl Method (ISWPropertySheet)

Gets the ActiveX control on the property sheet.

.NET Syntax

Visual Basic (Declaration)

```
Function GetControl( _  
    ByVal PageIndex As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISWPropertySheet  
Dim PageIndex As System.Integer  
Dim value As System.Object  
  
value = instance.GetControl(PageIndex)
```

C#

```
System.object GetControl(  
    System.int PageIndex  
)
```

C++/CLI

```
System.Object^ GetControl(  
& System.int PageIndex  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PageIndex
Index of property sheet

Return Value

ActiveX control

Visual Basic for Applications (VBA) Syntax

See [ISWPropertySheet::GetControl](#).

Remarks

Typically, you would call this method from the ISWPropertySheet [OnOKNotify](#) event handler to retrieve data from your ActiveX control.

See Also

[ISWPropertySheet Interface](#)
[ISWPropertySheet Members](#)

Availability

SOLIDWORKS 2004 SP4, Revision Number 12.4

SOLIDWORKS API Help

[Send comments on this topic.](#)

RemovePage Method (ISWPropertySheet)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISWPropertySheet Interface](#) : RemovePage Method (ISWPropertySheet)

Removes a CPropertyPage derived page from an [ISWPropertySheet](#).

.NET Syntax

Visual Basic (Declaration)

```
Function RemovePage( _  
    ByVal Page As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISWPropertySheet  
Dim Page As System.Integer  
Dim value As System.Integer  
  
value = instance.RemovePage(Page)
```

C#

```
System.int RemovePage(  
    System.int Page  
)
```

C++/CLI

```
System.int RemovePage(  
&    System.int Page  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Page
CPropertyPage to add, cast to a long

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SWPropertySheet::RemovePage](#).

See Also

[ISWPropertySheet Interface](#)
[ISWPropertySheet Members](#)

ISwScene (Type)

ISwScene Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISwScene Interface

Allows access to the scene of a model.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISwScene
```

Visual Basic (Usage)



```
Dim instance As ISwScene
```

C#

```
public interface ISwScene
```

C++/CLI

```
public interface class ISwScene
```

Visual Basic for Applications (VBA) Syntax

See [SwScene](#).

Example

[Get and Set Scene Properties \(VBA\)](#)
[Get and Set Scene Properties \(VB.NET\)](#)
[Get and Set Scene Properties \(C#\)](#)

Remarks

Call [IModelView::GraphicsRedraw](#) to repaint the graphics area after changing any scene.

Accessors

[IConfiguration::GetScene](#)

Access Diagram

[SwScene](#)

See Also

[ISwScene Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISwScene Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISwScene Interface

The following tables list the members exposed by [ISwScene](#).

Public Properties

Name	Description
 AccurateEnvLighting	Gets or sets whether to use the accurate environment lighting calculations from the high dynamic range imaging (HDRI) scene environment.
 AspectRatio	Gets the aspect ratio of the scene floor.
 BackgroundBottomGradientColor	Gets or sets the bottom color of the graduated range of background colors of this scene.
 BackgroundBrightness	Gets or sets the brightness of the background.
 BackgroundEnvImage	Gets or sets the environment image of this scene.
 BackgroundImage	Gets or sets the background image for this scene.
 BackgroundTopGradientColor	Gets or sets the background color of this scene.
 BackgroundType	Gets or sets the type of background for this scene.
 EnvironmentRotation	Gets or sets the rotation of the environment image of this scene.
 EnvironmentSize	Gets or sets the size of the high dynamic range imaging (HDRI) scene sphere that surrounds the model.
 FitToSWWindow	Gets or sets whether to stretch an image to fit the SOLIDWORKS window.
 FixedAspectRatio	Gets or sets whether to fix the aspect ratio of the scene floor.
 FlattenFloor	Get or sets whether to flatten the floor of a spherical environment to improve the look of models that naturally rest on flat floors.
 FloorAlignment	Gets or sets the plane with which to align the scene floor.
 FloorAutoSize	Gets or sets whether to resize the scene floor based on the model bounding box.
 FloorDepth	Gets or sets the depth of the scene floor.
 FloorDirection	Gets or sets whether to flip the direction of the floor of this scene.
 FloorOffset	Gets or sets the height of the model above or below the floor of this scene.
 FloorOffsetDirection	Gets or sets whether to swap the positions of the scene floor and the model.
 FloorReflections	Gets or sets whether to show reflections of the model on the scene floor.
 FloorRotation	Gets or sets the rotation of the scene floor relative to the environment of this scene.
 FloorSelection	Gets or sets the floor appearance of this scene.
 FloorShadows	Gets or sets whether to show shadows cast by the model on the scene floor.
 FloorWidth	Gets or sets the width of the scene floor.
 HorizonHeight	Gets or sets the height on the high dynamic range imaging (HDRI) scene sphere where the scene floor starts to flatten.
 KeepBackground	Gets or sets whether to retain the background when replacing the scene.
 RenderingBrightness	Gets or sets the brightness contributed by the high dynamic range imaging (HDRI) environment.
 SceneReflectivity	Gets or sets the amount of reflectivity provided by the high dynamic range imaging (HDRI) environment.

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Public Methods

Name	Description
 AddOrEditFloorAppearance	Adds or modifies the floor appearance of this scene.
 DeleteFloorAppearance	Deletes the floor appearance of this scene.
 GetFloorAppearance	Gets the floor appearance of this scene.
 GetFloorNormal	Gets the normal to the floor of this scene.
 GetP2SFileName	Gets the scene file for this scene.
 Modify	Modifies and saves this scene.
 OffsetToGeometry	Resets the floor offset.

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See Also

ISwScene Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISwScene Interface

For a list of all members of this type, see [ISwScene members](#).

Public Methods

Name	Description
 AddOrEditFloorAppearance	Adds or modifies the floor appearance of this scene.
 DeleteFloorAppearance	Deletes the floor appearance of this scene.
 GetFloorAppearance	Gets the floor appearance of this scene.
 GetFloorNormal	Gets the normal to the floor of this scene.
 GetP2SFileName	Gets the scene file for this scene.
 Modify	Modifies and saves this scene.
 OffsetToGeometry	Resets the floor offset.

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See Also

[ISwScene Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISwScene Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISwScene Interface

For a list of all members of this type, see [ISwScene members](#).

Public Properties

Name	Description
 AccurateEnvLighting	Gets or sets whether to use the accurate environment lighting calculations from the high dynamic range imaging (HDRI) scene environment.
 AspectRatio	Gets the aspect ratio of the scene floor.
 BackgroundBottomGradientColor	Gets or sets the bottom color of the graduated range of background colors of this scene.
 BackgroundBrightness	Gets or sets the brightness of the background.
 BackgroundEnvImage	Gets or sets the environment image of this scene.
 BackgroundImage	Gets or sets the background image for this scene.
 BackgroundTopGradientColor	Gets or sets the background color of this scene.
 BackgroundType	Gets or sets the type of background for this scene.
 EnvironmentRotation	Gets or sets the rotation of the environment image of this scene.
 EnvironmentSize	Gets or sets the size of the high dynamic range imaging (HDRI) scene sphere that surrounds the model.
 FitToSWWindow	Gets or sets whether to stretch an image to fit the SOLIDWORKS window.
 FixedAspectRatio	Gets or sets whether to fix the aspect ratio of the scene floor.
 FlattenFloor	Get or sets whether to flatten the floor of a spherical environment to improve the look of models that naturally rest on flat floors.
 FloorAlignment	Gets or sets the plane with which to align the scene floor.
 FloorAutoSize	Gets or sets whether to resize the scene floor based on the model bounding box.
 FloorDepth	Gets or sets the depth of the scene floor.

	FloorDirection	Gets or sets whether to flip the direction of the floor of this scene.
	FloorOffset	Gets or sets the height of the model above or below the floor of this scene.
	FloorOffsetDirection	Gets or sets whether to swap the positions of the scene floor and the model.
	FloorReflections	Gets or sets whether to show reflections of the model on the scene floor.
	FloorRotation	Gets or sets the rotation of the scene floor relative to the environment of this scene.
	FloorSelection	Gets or sets the floor appearance of this scene.
	FloorShadows	Gets or sets whether to show shadows cast by the model on the scene floor.
	FloorWidth	Gets or sets the width of the scene floor.
	HorizonHeight	Gets or sets the height on the high dynamic range imaging (HDRI) scene sphere where the scene floor starts to flatten.
	KeepBackground	Gets or sets whether to retain the background when replacing the scene.
	RenderingBrightness	Gets or sets the brightness contributed by the high dynamic range imaging (HDRI) environment.
	SceneReflectivity	Gets or sets the amount of reflectivity provided by the high dynamic range imaging (HDRI) environment.

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 See Also

[ISwScene Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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AccurateEnvLighting Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : AccurateEnvLighting Property (ISwScene)

Gets or sets whether to use the accurate environment lighting calculations from the high dynamic range imaging (HDRI) scene environment.

 .NET Syntax

Visual Basic (Declaration)

Property AccurateEnvLighting As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.AccurateEnvLighting = value
value = instance.AccurateEnvLighting
```

C#

```
System.bool AccurateEnvLighting {get; set;}
```

C++/CLI

```
property System.bool AccurateEnvLighting {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to use the accurate environment lighting calculations from the HDRI scene environment, false to not

 Visual Basic for Applications (VBA) Syntax

See [SwScene::AccurateEnvLighting](#).

Remarks

This property is valid only when a ray-trace rendering engine is active.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddOrEditFloorAppearance Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : AddOrEditFloorAppearance Method (ISwScene)

Adds or modifies the floor appearance of this scene.

.NET Syntax

Visual Basic (Declaration)

```
Sub AddOrEditFloorAppearance( _  
    ByVal Appearance As RenderMaterial _  
)
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim Appearance As RenderMaterial  
  
instance.AddOrEditFloorAppearance(Appearance)
```

C#

```
void AddOrEditFloorAppearance(  
    RenderMaterial Appearance  
)
```

C++/CLI

```
void AddOrEditFloorAppearance(  
& RenderMaterial^ Appearance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Appearance
[IRenderMaterial](#)

Visual Basic for Applications (VBA) Syntax

See [SwScene::AddOrEditFloorAppearance](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

[Send comments on this topic.](#)

AspectRatio Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : AspectRatio Property (ISwScene)

Gets the aspect ratio of the scene floor.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property AspectRatio As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Double
```

```
value = instance.AspectRatio
```

C#

```
System.double AspectRatio {get;}
```

C++/CLI

```
property System.double AspectRatio {  
    System.double get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Aspect ratio

Visual Basic for Applications (VBA) Syntax

See [SwScene::AspectRatio](#).

Remarks

To change the aspect ratio, set [ISwScene::FixedAspectRatio](#) to false and specify:

- [ISwScene::FloorDepth](#)
- [ISwScene::FloorWidth](#)

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BackgroundBottomGradientColor Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundBottomGradientColor Property (ISwScene)

Gets or sets the bottom color of the graduated range of background colors of this scene.

.NET Syntax

Visual Basic (Declaration)

```
Property BackgroundBottomGradientColor As System.Integer
```

Visual Basic (Usage)

```
Dim instance As ISwScene  
Dim value As System.Integer  
  
instance.BackgroundColor = value  
  
value = instance.BackgroundColor
```

C#

```
System.int BackgroundBottomGradientColor {get; set;}
```

C++/CLI

```
property System.int BackgroundBottomGradientColor {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

RGB color

Visual Basic for Applications (VBA) Syntax

See [SwScene::BackgroundBottomGradientColor](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is [swSceneBackgroundType_e.swBackgroundType_Graduated](#).
Set the top color of the graduated range of background colors with [ISwScene::BackgroundTopGradientColor](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

BackgroundBrightness Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundBrightness Property (ISwScene)

Gets or sets the brightness of the background.

.NET Syntax

Visual Basic (Declaration)

```
Property BackgroundBrightness As System.Double
```

Visual Basic (Usage)

```
Dim instance As ISwScene  
Dim value As System.Double  
  
instance.BackgroundBrightness = value
```

```
value = instance.BackgroundBrightness
```

C#

```
System.double BackgroundBrightness {get; set;}
```

C++/CLI

```
property System.double BackgroundBrightness {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Brightness

Visual Basic for Applications (VBA) Syntax

See [SwScene::BackgroundBrightness](#).

Remarks

This property is valid only when:

- a ray-trace rendering engine is activated
- [ISwScene::BackgroundType](#) is not [swSceneBackgroundType_e.swBackgroundType_None](#)

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

BackgroundEnvImage Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundEnvImage Property (ISwScene)

Gets or sets the environment image of this scene.

L.NET Syntax

Visual Basic (Declaration)

```
Property BackgroundEnvImage As System.String
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.String
```

```
instance.BackgroundEnvImage = value
```

```
value = instance.BackgroundEnvImage
```

C#

```
System.string BackgroundEnvImage {get; set;}
```

C++/CLI

```
property System.String^ BackgroundEnvImage {  
    System.String^ get();
```

```
void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Fully qualified path to an environment image file

Visual Basic for Applications (VBA) Syntax

See [ISwScene::BackgroundEnvImage](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is [swSceneBackgroundType_e.swBackgroundType_UseEnvironment](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BackgroundImage Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundImage Property (ISwScene)

Gets or sets the background image for this scene.

.NET Syntax

Visual Basic (Declaration)

Property BackgroundImage As System.String

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.String  
  
instance.BackgroundImage = value  
  
value = instance.BackgroundImage
```

C#

```
System.string BackgroundImage {get; set;}
```

C++/CLI

```
property System.String^ BackgroundImage {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Fully qualified path to a background image file

Visual Basic for Applications (VBA) Syntax

See [SwScene::BackgroundImage](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is [swSceneBackgroundType_e.swBackgroundType_Image](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)
[IModelDoc2::InsertBkgImage Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BackgroundTopGradientColor Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundTopGradientColor Property (ISwScene)

Gets or sets the background color of this scene.

.NET Syntax

Visual Basic (Declaration)

```
Property BackgroundTopGradientColor As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Integer  
  
instance.BackgroundTopGradientColor = value  
  
value = instance.BackgroundTopGradientColor
```

C#

```
System.int BackgroundTopGradientColor {get; set;}
```

C++/CLI

```
property System.int BackgroundTopGradientColor {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

RGB color

Visual Basic for Applications (VBA) Syntax

See [SwScene::BackgroundTopGradientColor](#).

Example

See the [ISwScene](#) examples.

Remarks

If ISwScene::BackgroundType is...	Then this property sets...
swSceneBackgroundType_e.swBackgroundType_Plain	a single background color.
swSceneBackgroundType_e.swBackgroundType_Graduated	the top color in the graduated range of background colors; Use ISwScene::BackgroundBottomGradientColor to set the bottom color in the graduated range of background colors.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments](#) on this topic.

BackgroundType Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : BackgroundType Property (ISwScene)

Gets or sets the type of background for this scene.

.NET Syntax

Visual Basic (Declaration)

Property BackgroundType As System.Integer

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Integer

instance.BackgroundType = value
value = instance.BackgroundType
```

C#

```
System.int BackgroundType {get; set;}
```

C++/CLI

```
property System.int BackgroundType {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of background as defined in [swSceneBackgroundType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwScene::BackgroundType](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

DeleteFloorAppearance Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : DeleteFloorAppearance Method (ISwScene)

Deletes the floor appearance of this scene.

.NET Syntax

Visual Basic (Declaration)

Function DeleteFloorAppearance() As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean
```

```
value = instance.DeleteFloorAppearance()
```

C#

```
System.bool DeleteFloorAppearance()
```

C++/CLI

```
System.bool DeleteFloorAppearance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [SwScene.DeleteFloorAppearance](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

EnvironmentRotation Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : EnvironmentRotation Property (ISwScene)

Gets or sets the rotation of the environment image of this scene.

.NET Syntax

Visual Basic (Declaration)

Property EnvironmentRotation As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.EnvironmentRotation = value

value = instance.EnvironmentRotation
```

C#

```
System.double EnvironmentRotation {get; set;}
```

C++/CLI

```
property System.double EnvironmentRotation {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

0.0 <= Rotation in degrees <= 360.0

Visual Basic for Applications (VBA) Syntax

See [SwScene::EnvironmentRotation](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

EnvironmentSize Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : EnvironmentSize Property (ISwScene)

Gets or sets the size of the high dynamic range imaging (HDRI) scene sphere that surrounds the model.

.NET Syntax

Visual Basic (Declaration)

```
Property EnvironmentSize As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.EnvironmentSize = value

value = instance.EnvironmentSize
```

C#

```
System.double EnvironmentSize {get; set;}
```

C++/CLI

```
property System.double EnvironmentSize {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Size of the HDRI scene sphere

Visual Basic for Applications (VBA) Syntax

See [ISwScene::EnvironmentSize](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::FlattenFloor](#) is set to true.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FitToSWWindow Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FitToSWWindow Property (ISwScene)

Gets or sets whether to stretch an image to fit the SOLIDWORKS window.

.NET Syntax

Visual Basic (Declaration)

```
Property FitToSWWindow As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FitToSWWindow = value
value = instance.FitToSWWindow
```

C#

```
System.bool FitToSWWindow {get; set;}
```

C++/CLI

```
property System.bool FitToSWWindow {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to stretch an image to fit, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FitToSWWindow](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is [swSceneBackgroundType_e.swBackgroundType_Image](#).

Set the background image with [ISwScene::BackgroundImage](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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FixedAspectRatio Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FixedAspectRatio Property (ISwScene)

Gets or sets whether to fix the aspect ratio of the scene floor.

.NET Syntax

Visual Basic (Declaration)

Property FixedAspectRatio As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FixedAspectRatio = value

value = instance.FixedAspectRatio
```

C#

```
System.bool FixedAspectRatio {get; set;}
```

C++/CLI

```
property System.bool FixedAspectRatio {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to fix the aspect ratio of the scene floor, false to allow the scene floor width and height to change independently

Visual Basic for Applications (VBA) Syntax

See [SwScene::FixedAspectRatio](#).

Example

See the [ISwScene](#) examples.

Remarks

If this property is set to true, then you need only specify one of the following:

- [ISwScene::FloorDepth](#)
- [ISwScene::FloorWidth](#)

To change the [aspect ratio](#), set this property to false and set both of the following:

- [ISwScene::FloorDepth](#)
- [ISwScene::FloorWidth](#)

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

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FlattenFloor Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FlattenFloor Property (ISwScene)

Get or sets whether to flatten the floor of a spherical environment to improve the look of models that naturally rest on flat floors.

.NET Syntax

Visual Basic (Declaration)

Property FlattenFloor As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FlattenFloor = value
value = instance.FlattenFloor
```

C#

```
System.bool FlattenFloor {get; set;}
```

C++/CLI

```
property System.bool FlattenFloor {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to flatten the floor, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FlattenFloor](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is [swSceneBackgroundType_e.swBackgroundType_UseEnvironment](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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FloorAlignment Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorAlignment Property (ISwScene)

Gets or sets the plane with which to align the scene floor.

.NET Syntax

Visual Basic (Declaration)

```
Property FloorAlignment As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Integer  
  
instance.FloorAlignment = value  
  
value = instance.FloorAlignment
```

C#

```
System.int FloorAlignment {get; set;}
```

C++/CLI

```
property System.int FloorAlignment {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Scene floor alignment plane as defined by [swSceneFloorAlign_e](#)

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorAlignment](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

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FloorAutoSize Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorAutoSize Property (ISwScene)

Gets or sets whether to resize the scene floor based on the model bounding box.

.NET Syntax

Visual Basic (Declaration)

Property FloorAutoSize As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FloorAutoSize = value

value = instance.FloorAutoSize
```

C#

```
System.bool FloorAutoSize {get; set;}
```

C++/CLI

```
property System.bool FloorAutoSize {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to resize the scene floor based on the model bounding box, false to manually specify the width, depth, and rotation of the scene floor

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorAutoSize](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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FloorDepth Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorDepth Property (ISwScene)

Gets or sets the depth of the scene floor.

.NET Syntax

Visual Basic (Declaration)

Property FloorDepth As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.FloorDepth = value

value = instance.FloorDepth
```

C#

```
System.double FloorDepth {get; set;}
```

C++/CLI

```
property System.double FloorDepth {
    System.double get();
    void set ( &    System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Depth of the scene floor

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorDepth](#).

Remarks

The setting of this property is valid only if [ISwScene::FloorAutoSize](#) is false.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

FloorDirection Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorDirection Property (ISwScene)

Gets or sets whether to flip the direction of the floor of this scene.

.NET Syntax

Visual Basic (Declaration)

```
Property FloorDirection As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FloorDirection = value

value = instance.FloorDirection
```

C#

```
System.bool FloorDirection {get; set;}
```

C++/CLI

```
property System.bool FloorDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to flip the direction of the floor, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorDirection](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

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FloorOffset Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorOffset Property (ISwScene)

Gets or sets the height of the model above or below the floor of this scene.

.NET Syntax

Visual Basic (Declaration)

Property FloorOffset As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.FloorOffset = value
value = instance.FloorOffset
```

C#

```
System.double FloorOffset {get; set;}
```

C++/CLI

```
property System.double FloorOffset {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Distance between the scene floor and the model

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorOffset](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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FloorOffsetDirection Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorOffsetDirection Property (ISwScene)

Gets or sets whether to swap the positions of the scene floor and the model.

.NET Syntax

Visual Basic (Declaration)

Property FloorOffsetDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean

instance.FloorOffsetDirection = value

value = instance.FloorOffsetDirection
```

C#

```
System.bool FloorOffsetDirection {get; set;}
```

C++/CLI

```
property System.bool FloorOffsetDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to swap positions of the scene floor and the model, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorOffsetDirection](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

FloorReflections Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorReflections Property (ISwScene)

Gets or sets whether to show reflections of the model on the scene floor.

.NET Syntax

Visual Basic (Declaration)

Property FloorReflections As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Boolean  
  
instance.FloorReflections = value  
value = instance.FloorReflections
```

C#

```
System.bool FloorReflections {get; set;}
```

C++/CLI

```
property System.bool FloorReflections {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to show reflections, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorReflections](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::FloorShadows](#) is set to true.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

[Send comments on this topic.](#)

FloorRotation Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorRotation Property (ISwScene)

Gets or sets the rotation of the scene floor relative to the environment of this scene.

.NET Syntax

Visual Basic (Declaration)

Property FloorRotation As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.FloorRotation = value

value = instance.FloorRotation
```

C#

```
System.double FloorRotation {get; set;}
```

C++/CLI

```
property System.double FloorRotation {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

0.0 <= Rotation in degrees <= 360.0

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorRotation](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::FloorAutoSize](#) is false.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

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FloorSelection Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorSelection Property (ISwScene)

Gets or sets the floor appearance of this scene.

.NET Syntax

Visual Basic (Declaration)

Property FloorSelection As System.Object

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Object  
  
instance.FloorSelection = value  
value = instance.FloorSelection
```

C#

```
System.object FloorSelection {get; set;}
```

C++/CLI

```
property System.Object^ FloorSelection {  
    System.Object^ get();  
    void set ( &    System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IAppearanceSetting](#)

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorSelection](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FloorShadows Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorShadows Property (ISwScene)

Gets or sets whether to show shadows cast by the model on the scene floor.

.NET Syntax

Visual Basic (Declaration)

Property FloorShadows As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Boolean  
  
instance.FloorShadows = value  
value = instance.FloorShadows
```

C#

```
System.bool FloorShadows {get; set;}
```

C++/CLI

```
property System.bool FloorShadows {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to show shadows cast by the model, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorShadows](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SOLIDWORKS API Help

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FloorWidth Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : FloorWidth Property (ISwScene)

Gets or sets the width of the scene floor.

.NET Syntax

Visual Basic (Declaration)

Property FloorWidth As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.FloorWidth = value
value = instance.FloorWidth
```

C#

```
System.double FloorWidth {get; set;}
```

C++/CLI

```
property System.double FloorWidth {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Width of the scene floor

Visual Basic for Applications (VBA) Syntax

See [SwScene::FloorWidth](#).

 Remarks

The setting of this property is valid only if [ISwScene::FloorAutoSize](#) is false.

 See Also

[ISwScene Interface](#)
[ISwScene Members](#)

 Availability

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GetFloorAppearance Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : GetFloorAppearance Method (ISwScene)

Gets the floor appearance of this scene.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetFloorAppearance() As RenderMaterial
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As RenderMaterial  
  
value = instance.GetFloorAppearance()
```

C#

```
RenderMaterial GetFloorAppearance()
```

C++/CLI

```
RenderMaterial^ GetFloorAppearance();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IRenderMaterial](#)

 Visual Basic for Applications (VBA) Syntax

See [SwScene::GetFloorAppearance](#).

 See Also

[ISwScene Interface](#)
[ISwScene Members](#)

 Availability

SOLIDWORKS 2025 FCS, Revision Number 33

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GetFloorNormal Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : GetFloorNormal Method (ISwScene)

Gets the normal to the floor of this scene.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetFloorNormal( _  
    ByRef Point As MathPoint, _  
    ByRef Normal As MathVector _  
)
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim Point As MathPoint  
Dim Normal As MathVector  
  
instance.GetFloorNormal(Point, Normal)
```

C#

```
void GetFloorNormal(  
    out MathPoint Point,  
    out MathVector Normal  
)
```

C++/CLI

```
void GetFloorNormal(  
& [Out] MathPoint^ Point,  
& [Out] MathVector^ Normal  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Point
[IMathPoint](#)
Normal
[IMathVector](#)

Visual Basic for Applications (VBA) Syntax

See [SwScene::GetFloorNormal](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

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GetP2SFileName Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : GetP2SFileName Method (ISwScene)

Gets the scene file for this scene.

.NET Syntax

Visual Basic (Declaration)

```
Sub GetP2SFileName( _  
    ByRef Val As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim Val As System.String  
  
instance.GetP2SFileName(Val)
```

C#

```
void GetP2SFileName(  
    out System.string Val  
)
```

C++/CLI

```
void GetP2SFileName(  
& [Out] System.String^ Val  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Val
Fully qualified path to a scene file (*.p2s)

Visual Basic for Applications (VBA) Syntax

See [SwScene::GetP2SFileName](#).

Example

See the [ISwScene](#) examples.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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HorizonHeight Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : HorizonHeight Property (ISwScene)

Gets or sets the height on the high dynamic range imaging (HDRI) scene sphere where the scene floor starts to flatten.

.NET Syntax

Visual Basic (Declaration)

```
Property HorizonHeight As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Double  
  
instance.HorizonHeight = value
```

```
value = instance.HorizonHeight
```

C#

```
System.double HorizonHeight {get; set;}
```

C++/CLI

```
property System.double HorizonHeight {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

0.0 <= Horizon height <= 1.0

Visual Basic for Applications (VBA) Syntax

See [SwScene::HorizonHeight](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::FlattenFloor](#) is set to true.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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KeepBackground Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : KeepBackground Property (ISwScene)

Gets or sets whether to retain the background when replacing the scene.

.NET Syntax

Visual Basic (Declaration)

```
Property KeepBackground As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Boolean  
  
instance.KeepBackground = value  
  
value = instance.KeepBackground
```

C#

```
System.bool KeepBackground {get; set;}
```

C++/CLI

```
property System.bool KeepBackground {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to retain the background when replacing a scene, false to not

Visual Basic for Applications (VBA) Syntax

See [SwScene::KeepBackground](#).

Example

See the [ISwScene](#) examples.

Remarks

This property is valid only if [ISwScene::BackgroundType](#) is set to one of the following:

- [swSceneBackgroundType_e.swBackgroundType_Image](#)
- [swSceneBackgroundType_e.swBackgroundType_Graduated](#)
- [swSceneBackgroundType_e.swBackgroundType_Plain](#)

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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Modify Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : Modify Method (ISwScene)

Modifies and saves this scene.

.NET Syntax

Visual Basic (Declaration)

```
Function Modify() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Boolean
```

```
value = instance.Modify()
```

C#

```
System.bool Modify()
```

C++/CLI

```
System.bool Modify();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the scene is successfully modified and saved, false if not

Visual Basic for Applications (VBA) Syntax

See [ISwScene::Modify](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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[Send comments on this topic.](#)

OffsetToGeometry Method (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : OffsetToGeometry Method (ISwScene)

Resets the floor offset.

.NET Syntax

Visual Basic (Declaration)

```
Function OffsetToGeometry() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Boolean  
  
value = instance.OffsetToGeometry()
```

C#

```
System.bool OffsetToGeometry()
```

C++/CLI

```
System.bool OffsetToGeometry();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ISwScene::OffsetToGeometry](#).

Remarks

Calling this method invalidates previous calls to [ISwScene::FloorOffset](#) and [ISwScene::FloorOffsetDirection](#).

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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RenderingBrightness Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : RenderingBrightness Property (ISwScene)

Gets or sets the brightness contributed by the high dynamic range imaging (HDRI) environment.

.NET Syntax

Visual Basic (Declaration)

Property RenderingBrightness As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene
Dim value As System.Double

instance.RenderingBrightness = value
value = instance.RenderingBrightness
```

C#

```
System.double RenderingBrightness {get; set;}
```

C++/CLI

```
property System.double RenderingBrightness {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Brightness

Visual Basic for Applications (VBA) Syntax

See [SwScene::RenderingBrightness](#).

Remarks

This property is valid only when a ray-trace rendering engine is activated.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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SceneReflectivity Property (ISwScene)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISwScene Interface](#) : SceneReflectivity Property (ISwScene)

Gets or sets the amount of reflectivity provided by the high dynamic range imaging (HDRI) environment.

.NET Syntax

Visual Basic (Declaration)

Property SceneReflectivity As System.Double

Visual Basic (Usage)



```
Dim instance As ISwScene  
Dim value As System.Double  
  
instance.SceneReflectivity = value  
value = instance.SceneReflectivity
```

C#

```
System.double SceneReflectivity {get; set;}
```

C++/CLI

```
property System.double SceneReflectivity {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Reflectivity

Visual Basic for Applications (VBA) Syntax

See [SwScene::SceneReflectivity](#).

Remarks

This property is valid only when a ray-trace rendering engine is activated.

See Also

[ISwScene Interface](#)
[ISwScene Members](#)

Availability

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ISymmetricMateFeatureData (Type)

SOLIDWORKS API Help

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ISymmetricMateFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISymmetricMateFeatureData Interface

Allows access to symmetry mate feature data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface ISymmetricMateFeatureData

Visual Basic (Usage)



Dim instance As [ISymmetricMateFeatureData](#)

C#

public interface ISymmetricMateFeatureData

C++/CLI

public interface class ISymmetricMateFeatureData

Visual Basic for Applications (VBA) Syntax

See [SymmetricMateFeatureData](#).

Example

[Create and Edit Symmetric Mate \(VBA\)](#)
[Create and Edit Symmetric Mate \(C#\)](#)

Remarks

Symmetry mates force two similar entities to be symmetric about a plane or planar face.

[IMateFeatureData](#) is the parent of this advanced mate interface.

To create a symmetry mate:

1. Follow general instructions in the [IAssemblyDoc::CreateMate](#) Remarks.
2. Specify [ISymmetricMateFeatureData::EntitiesToMate](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the entities using Mark = 1.
3. Specify [ISymmetricMateFeatureData::SymmetryPlane](#) or use [IModelDocExtension::SelectByID2](#) to pre-select the symmetry plane using Mark = 4.
4. Specify other properties of the SymmetricMateFeatureData object as required.

To edit a symmetry mate:

1. Access its feature and call [IFeature::GetDefinition](#) to get the MateFeatureData object.
2. Cast the MateFeatureData object to a SymmetricMateFeatureData object.
3. Modify the SymmetricMateFeatureData object.
4. Call [IFeature::ModifyDefinition](#).

To delete a symmetry mate, use [IModelDocExtension::DeleteSelection2](#).

Access Diagram

[SymmetricMateFeatureData](#)

See Also

[ISymmetricMateFeatureData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

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ISymmetricMateFeatureData Interface Members

[See Also](#) [Properties](#)



The following tables list the members exposed by [ISymmetricMateFeatureData](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities in this symmetry mate.
 MateAlignment	Gets or sets the alignment of this symmetry mate.
 SymmetryPlane	Gets or sets the symmetry plane of this symmetry mate.

[Top](#)

See Also

[ISymmetricMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISymmetricMateFeatureData Interface Properties

[See Also](#)

SolidWorks.Interop.sldworks Namespace : ISymmetricMateFeatureData Interface

For a list of all members of this type, see [ISymmetricMateFeatureData members](#).

Public Properties

Name	Description
 EntitiesToMate	Gets or sets the entities in this symmetry mate.
 MateAlignment	Gets or sets the alignment of this symmetry mate.
 SymmetryPlane	Gets or sets the symmetry plane of this symmetry mate.

[Top](#)

See Also

[ISymmetricMateFeatureData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

EntitiesToMate Property (ISymmetricMateFeatureData)

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) > [ISymmetricMateFeatureData Interface](#) : EntitiesToMate Property (ISymmetricMateFeatureData)

Gets or sets the entities in this symmetry mate.

.NET Syntax

Visual Basic (Declaration)

Property EntitiesToMate As System.Object

Visual Basic (Usage)

 Copy Code

Dim instance As [ISymmetricMateFeatureData](#)
Dim value As System.Object

instance.EntitiesToMate = value

value = instance.EntitiesToMate

C#

System.object EntitiesToMate {get; set;}

C++/CLI

```
property System.Object^ EntitiesToMate {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of two similar entities: [vertexes](#), [sketch points](#), [edges](#), [axes](#), [sketch lines](#), [planes](#), [planar faces](#), spheres of equal radii, or cylinders of equal radii

Visual Basic for Applications (VBA) Syntax

See [SymmetricMateFeatureData.EntitiesToMate](#).

Example

See the [ISymmetricMateFeatureData](#) examples.

See Also

[ISymmetricMateFeatureData Interface](#)
[ISymmetricMateFeatureData Members](#)

Availability

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MateAlignment Property (ISymmetricMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISymmetricMateFeatureData Interface](#) : MateAlignment Property (ISymmetricMateFeatureData)

Gets or sets the alignment of this symmetry mate.

.NET Syntax

Visual Basic (Declaration)

```
Property MateAlignment As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISymmetricMateFeatureData
Dim value As System.Integer
```

```
instance.MateAlignment = value
```

```
value = instance.MateAlignment
```

C#

```
System.int MateAlignment {get; set;}
```

C++/CLI

```
property System.int MateAlignment {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Mate alignment as defined in [swMateAlign_e](#)

Visual Basic for Applications (VBA) Syntax

See [SymmetricMateFeatureData::MateAlignment](#).

Example

See the [ISymmetricMateFeatureData](#) examples.

Remarks

If this property is set to `swMateAlign_e`:

- `swMateAlignALIGNED`, then vectors normal to the selected entities point in the same direction.
- `swMateAlignANTI_ALIGNED`, then vectors normal to the selected entities point in opposite directions.

See Also

[ISymmetricMateFeatureData Interface](#)
[ISymmetricMateFeatureData Members](#)

Availability

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SymmetryPlane Property (ISymmetricMateFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISymmetricMateFeatureData Interface](#) : SymmetryPlane Property (ISymmetricMateFeatureData)

Gets or sets the symmetry plane of this symmetry mate.

.NET Syntax

Visual Basic (Declaration)

Property SymmetryPlane As System.Object

Visual Basic (Usage)



```
Dim instance As ISymmetricMateFeatureData  
Dim value As System.Object
```

```
instance.SymmetryPlane = value
```

```
value = instance.SymmetryPlane
```

C#

```
System.object SymmetryPlane {get; set;}
```

C++/CLI

```
property System.Object^ SymmetryPlane {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Front, Top, or Right plane or [IRefPlane](#)

Visual Basic for Applications (VBA) Syntax

See [SymmetricMateFeatureData::SymmetryPlane](#).

Example

See the [ISymmetricMateFeatureData](#) examples.

See Also

[ISymmetricMateFeatureData Interface](#)
[ISymmetricMateFeatureData Members](#)

Availability

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